

Double-Inertia, Taylor Rules, and Monetary Policy Gradualism

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Abstract

Empirically estimated Taylor rules are often “inertial”: they contain the lag of the policy rate to capture gradualism in monetary policy. This paper revisits the “double-inertial” Taylor rule proposed by [Carlstrom and Fuerst \(2014\)](#) that gradually adjusts *changes* in the policy rate rather than just its level. The double-inertial rule explains more than twice the variation of changes in the federal funds rate relative to the standard inertial Taylor rule in recent decades. In fact, an AR(2) model for the policy rate—which is equivalent to a double-inertial Taylor rule that ignores current macroeconomic conditions—predicts changes in the federal funds rate virtually just as well at short horizons, highlighting the quantitative importance of inertia in both the level and rate of change of the policy rate. However, without incorporating economic fundamentals, the AR(2) model struggles to explain the trajectory of the federal funds rate over longer horizons.

1 Introduction

Since their conception in the early 1990s, Taylor rules have been one of the most influential frameworks for describing monetary policy ([Taylor, 1993](#)). Despite their simplicity, these rules were immediately recognized for their remarkable ability to explain the historical trajectory of the federal funds rate through deviations in inflation and economic activity from their potential levels. A growing literature interested in the descriptive capabilities of these rules quickly centered on another essential observation: that adding the lag of the policy rate greatly improves the fit of these rules, especially in the United States and following Paul Volcker’s tenure as Chair of the Federal Reserve.

The success of these “inertial” Taylor rules, which are now widely adopted in reduced-form and structural macroeconomic models, is commonly thought to be a reflection of policymakers’ desire to avoid large, abrupt changes in the level of the policy rate and instead to move the policy rate *gradually* in response to economic fundamentals. This characteristic of monetary policy has been reinforced and explained both in the academic literature and through the communications of policymakers with the public (e.g. [Clarida](#)

et al., 2000; Bernanke, 2004). Proponents of monetary policy “gradualism” have gone on to provide normative arguments, explaining inertia as reflecting, for example, an optimal response to uncertainty and volatility in the economy and financial markets (Woodford, 1999; Sack and Wieland, 2000; Stein and Sunderam, 2018; Levin et al., 1999).

Carlstrom and Fuerst (2014) introduced the notion of “double-inertia” in Taylor rules. They added a second lag of the policy rate to an inertial Taylor rule to capture the desire of policymakers to maintain a steady pace of change in the policy rate. Their observation was that central banks appear reluctant to reverse direction abruptly—shifting from tightening to easing or vice versa—and often adjust rates in sequences of similarly sized increments. Using data from 1993 to 2008, Carlstrom and Fuerst (2014) documented improved fit for this specification, a finding that corroborated previous work by Coibion and Gorodnichenko (2012), who estimated a version of a Taylor rule with two lags.

This paper provides a systematic empirical evaluation of the double-inertial Taylor rule that extends to 2025, thereby doubling the sample length of Carlstrom and Fuerst (2014) and Coibion and Gorodnichenko (2012). The sample includes two episodes during which the policy rate hit its effective lower bound (ELB). The eventual departures from the ELB may be particularly important data points for understanding inertial behavior.

We find strong evidence for the double-inertial rule in explaining the settings of the federal funds rate in recent decades. With an R^2 of 0.57, the double inertial rule is able to explain more than twice the variation in quarterly changes to the federal funds rate across the relevant sample than the best single-inertial rule. This result suggests that the standard practice of modeling monetary policy with a single-inertial rule in macroeconomic models, as is done commonly in academia and central banks alike, is a source of misspecification. Our result is robust to alternative data subsets, the use of shadow rate estimates during effective lower bound periods, substitution of an output gap measure for the unemployment gap, and the use of real-time rather than realized data in the spirit of Orphanides (2001).

The success of inertial rules may simply reflect the fact that first-order autoregressive models are often the best predictor of macroeconomic time series, including interest rates. Indeed, simple autoregressive models do virtually just as well in predicting one-quarter-ahead changes in the federal funds rate as single-inertial Taylor rules. However, at longer horizons, the Taylor rules, which employ economic fundamentals in predicting the policy rate, substantially outperform simple autoregressive benchmarks. Economic fundamentals thus contribute little to short-run predictions of the policy rate but are essential for predicting the path of the policy rate over longer horizons.

We also evaluate the empirical performance of inertial rules during the historical period starting in 1960 and ending with Volcker’s resignation. A large literature indicates that monetary policy was structurally different before the 1980s, exhibiting weaker responses to inflation expectations, more focus on monetary aggregates, less reliance on policy rules, and lower inertia (e.g. Clarida et al., 2000; Taylor, 1999). In line with this literature, we cannot reject the null that the coefficient on our double-inertia term is zero prior to 1987.

Our paper is empirical and our aim is not to evaluate whether it is desirable or optimal for monetary policy to act gradually. That said, Woodford (1999) derives the interest rate rule that implements optimal monetary policy in a standard New-Keynesian model and

shows that it features two lags in the policy rate, just like the double-inertial rule.

The structure of the rest of the paper is as follows. Section 2 presents our Taylor rules and benchmark models used in estimation. Section 3 describes our data and methodological approach. Section 4 evaluates our primary results and goodness of fit metrics. Section 5 details our additional robustness exercises. Section 6 concludes.

2 Monetary policy rules

In this section, we review Taylor rule specifications with and without inertia and revisit the double-inertial rule of [Carlstrom and Fuerst \(2014\)](#).

Let i_t be the nominal policy rate at time t . A non-inertial Taylor rule TR_t relates the nominal policy rate i_t to inflation and a measure of resource utilization in the economy. When the policy rate is not constrained by the effective lower bound (ELB), the policy rate is set according to:

$$i_t = \underbrace{r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - \tilde{u}_t)}_{=:TR_t}, \quad (1)$$

where r^* is the longer-run neutral real rate (assumed to be constant over a given data window), π_t is the inflation rate, u_t is the unemployment rate, π^* is the target rate of inflation, and $\tilde{u}_t$ is the natural rate of unemployment. In estimation, we account for the effective lower bound by specifying the latent policy rate as:

$$i_t^* = \theta_0 + \theta_1\pi_t + \theta_2(u_t - \tilde{u}_t) + \varepsilon_t, \quad (2)$$

and we then observe

$$i_t = \max\{\underline{i}, i_t^*\}, \quad (3)$$

where the max operator captures the constraint that the interest rate cannot fall below the ELB, $\underline{i}$.

The standard way of adding gradualism to this rule is to assume that the interest rate is set as a weighted average of the Taylor rule prescription, TR_t , and the previous period's interest rate. We thus introduce a smoothing parameter ρ that captures the degree of persistence in the nominal policy rate across periods. The resulting specification is the “single-inertial” Taylor rule. Away from the ELB, this rule prescribes:

$$i_t = \rho i_{t-1} + (1 - \rho)TR_t. \quad (4)$$

By subtracting i_{t-1} from both sides, this rule can be rewritten into a partial-adjustment specification that relates the pace of change in interest rates to the difference between the desired and the previous level of the policy rate:

$$\Delta i_t = (1 - \rho)(TR_t - i_{t-1}),$$

where $(1 - \rho)$ is the proportion of movement from the current level of the policy rate towards the intermediate rate target given by the non-inertial Taylor rule.

Expanding the levels specification of the inertial rule yields the following:

$$i_t = \rho i_{t-1} + (1 - \rho) [r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - \tilde{u}_t)]. \quad (5)$$

We recover the parameters of this rule from the estimation of the corresponding latent policy rate specification:

$$i_t = \max\{\underline{i}, \theta_0 + \theta_1\pi_t + \theta_2(u_t - \tilde{u}_t) + \theta_3i_{t-1} + \varepsilon_t\}. \quad (6)$$

Carlstrom and Fuerst (2014) introduced the “double-inertial” rule, which smooths not only the level of the policy rate, but also its pace of change. The motivating idea is that central banks are reluctant to abruptly go from a hike to a cut even when the cut is small, and often adjust the policy rate in a series of equal-size increments. This desire to smooth the pace of changes in the policy rate is captured with the addition of a parameter γ in the partial-adjustment specification:

$$\Delta i_t = \gamma \Delta i_{t-1} + (1 - \rho)(TR_t - i_{t-1}).$$

Expanding and converting into levels, the double-inertial Taylor rule away from the ELB takes the form:

$$i_t = -\gamma i_{t-2} + (\rho + \gamma)i_{t-1} + (1 - \rho) [r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - \tilde{u}_t)]. \quad (7)$$

We recover the parameters of this rule by estimating the corresponding latent policy rate specification:

$$i_t = \max\{\underline{i}, \theta_0 + \theta_1\pi_t + \theta_2(u_t - \tilde{u}_t) + \theta_3i_{t-1} + \theta_4i_{t-2} + \varepsilon_t\}. \quad (8)$$

To understand the extent to which the strong empirical performance of inertial Taylor rules comes simply from anchoring predictions for the current interest rate to their previous level, we also evaluate AR(1) and AR(2) benchmark models of the policy rate, where we set θ_1 and θ_2 to zero in (6) and (8).

3 Data

All of the data used in our primary estimation procedure is sourced from The Federal Reserve Bank of St. Louis’s FRED database. We focus on the 2025:Q3 data vintage, which allows us to estimate a longer sample than would be possible using real-time data, as in Orphanides (2001).¹ We extract data for the federal funds effective rate (FEDFUNDS), the personal consumption expenditures price index (PCEPILFE), the unemployment rate (UNRATE), and the natural rate of unemployment (NROU). We then construct an unemployment gap variable by computing the difference between the unemployment rate

¹Section 5 shows that our conclusions hold with real-time data.

and the natural rate of unemployment in every quarter. The inflation target π^* is not estimated but instead set to the Federal Reserve’s 2 percent goal. While this target was not formally announced until 2012, we treat it as a reasonable proxy for the low and stable inflation mandate that guided policy in earlier decades.

We focus our analysis on three data samples. The first sample runs from 1987:Q4 to 2006:Q4, which covers the period following Paul Volcker’s resignation as Chair of the Federal Reserve Board but prior to the events associated with the 2008 recession. This window corresponds fairly closely to the data used in previous analysis of Taylor rules with double-inertia, which also ends before the 2008 crisis. We estimate the parameters of our policy rules over this period using OLS regression in line with [Carvalho et al. \(2021\)](#), who show that the endogeneity bias is small.

The second sample additionally includes the period between 2007:Q1 and 2025:Q3. This extension doubles the sample size used in previous work, allowing us to reassess the prevalence of double-inertia in recent decades with greater precision. During this period, the Federal Reserve held the policy rate at the effective lower bound (ELB) from December 16, 2008 to December 15, 2015, and again from March 15, 2020 through March 16, 2022. As a result, our extended sample includes “liftoff” episodes, which may be especially important in identifying inertial behavior. Because changes in the stance of monetary policy are not reflected in the federal funds rate during ELB periods, we employ Tobit regression to estimate the rule parameters in our model of the latent policy rate.

Our third data subset runs from 1960:Q1 to 1987:Q3. This window provides a valuable comparison that both highlights the importance of double-inertia in prediction and helps to support its interpretation in the context of monetary policy gradualism. The literature has long recognized that Taylor rules—particularly those with inertia—describe monetary policy far more accurately in the post-Volcker era (e.g. [Clarida et al., 2000](#), [Taylor, 1999](#)). This observation is generally understood as reflecting substantial shifts in the Federal Reserve’s approach to monetary policy in the late eighties and early nineties, including stronger adherence to the Taylor principle, greater concern with market expectations, and a more gradual policy rate adjustment process. To the extent that our estimates suggest double-inertia is more pronounced in recent decades than it was in the historical sample, this provides evidence that it may be a characteristic feature of gradualist policy regimes.

4 Results

4.1 Rule parameters

Table 1 reports the estimated parameters of each Taylor rule variant.

For both the pre-2007 sample and the full post-1987 sample, the inflation and unemployment gap coefficients have the expected signs in all models. However, in the full sample, the inflation coefficient for the non-inertial Taylor rule is small with a large standard error. At 0.92, the implied policy response to inflation is less than one-for-one over the period, implying a violation of the Taylor principle. In contrast, the estimated inertial and double-inertial rules have inflation and unemployment gap coefficients that are

substantially greater than one in magnitude, though they are too imprecisely estimated to be labeled as significantly greater. In the pre-2007 sample, however, all three of the rules provide estimates of the inflation coefficient that are significantly greater than one at the 5 percent level.

The r^* estimates generated by the Taylor rule variants are very stable across models within the pre-2007 sample, clustering between 2.30 and 2.40. The estimates are considerably lower in the full sample, ranging from 0.32 for the double-inertial model to 1.27 for the standard Taylor rule. These levels are broadly consistent with r^* estimates over recent decades, and the differences across the samples likely reflect a decline in the natural rate during the late 2000s and early 2010s as well as the extended ELB periods.

Policy rate inertia, as captured by ρ , is large and highly significant in the pre-2007 sample. In the absence of double-inertia, these estimates imply partial adjustments of 13 and 20 percent toward the implied intermediate rate target per quarter. In the full sample, the estimated inertia is slightly higher, implying partial adjustments of less than 10 percent. That is, more than 90 percent of the level of the current federal funds rate in a given quarter can be attributed to its lagged value.

Most notably, the double-inertia coefficient γ is estimated fairly precisely at 0.61 and 0.63 in the pre-2007 and full samples, respectively. The estimates are highly significant, easily rejecting the null of the single-inertial model as sufficient in describing the dynamics of the policy rate. These results together corroborate previous findings that inertia, and hence gradualism, play a significant role in monetary policy through the 1990s and into the 2000s. In addition, they are evidence that gradualism encompasses persistence in changes to the policy rate: policymakers tend to pursue interest rate adjustments of similar size and in the same direction across periods. The strong evidence in favor of the double-inertial rule in the full sample underscores the continued relevance of change-smoothing in the monetary policy gradualism of recent decades.

4.2 Goodness of fit

To better quantify and evaluate the performance of each policy rule across subsets, we construct three R^2 statistic variants.

The first measure is standard, describing the explanatory power of each estimated rule with respect to the *level* of the policy rate, given realized values of the inflation and unemployment gaps in the next quarter. Let r index a particular policy rule and s index the data subset. Then, we can express this measure formally as follows:

$$R_r^2 = 1 - \frac{\sum_t (i_t - \hat{i}_{t,r})^2}{\sum_t (i_t - \bar{i})^2}, \quad (9)$$

where $\bar{i}$ is the average federal funds rate over the sample and $\hat{i}_{t,r}$ denotes the policy rate predicted by the estimated rule r in period t . Note that these policy rate predictions are truncated below at 0 to account for the ELB, as described in Section 2. We refer to this measure as “next-quarter levels” R^2 .

Table 1: Recovered Rule Parameters by Sample

Parameter	Pre-2007			Post-1987		
	Non-inertial	Inertial	Double-inertial	Non-inertial	Inertial	Double-inertial
$\hat{r}^*$	2.37 (0.29)	2.34 (0.46)	2.37 (0.38)	1.27 (0.37)	0.39 (1.17)	0.32 (0.95)
$\hat{\beta}_\pi$	1.68 (0.20)	1.74 (0.32)	1.51 (0.29)	0.92 (0.65)	2.38 (0.98)	1.88 (0.75)
$\hat{\beta}_u$	-1.73 (0.20)	-2.30 (0.34)	-1.66 (0.46)	-1.62 (0.33)	-4.34 (1.67)	-3.09 (1.12)
$\hat{\rho}$		0.80 (0.05)	0.87 (0.03)		0.94 (0.03)	0.96 (0.01)
$\hat{\gamma}$			0.61 (0.12)			0.63 (0.08)
N	77	77	77	152	152	152

Note: Estimated using quarterly data from FRED. Parameters $\hat{r}^*$, $\hat{\beta}_\pi$, $\hat{\beta}_u$, $\hat{\rho}$, and $\hat{\gamma}$ are recovered from regression coefficients, and Newey-West standard errors with a lag length of 4 are computed via the delta method. The 'Pre-2007' sample runs from 1987:Q4 to 2006:Q4, and is estimated with OLS. The full 'Post-1987' sample runs from 1987:Q4 to 2025:Q3, and is estimated with Tobit regression.

Across both samples, the pattern of results for this measure, as reported in the first and sixth columns of Table 2, is broadly similar. In the pre-2007 sample, the non-inertial rule achieves a moderate R^2 of 0.78, while the inertial and double-inertial rules are able to explain almost all of the variation in the policy rate, achieving R^2 values of 0.97 and 0.98, respectively. In the full post-1987 sample, this disparity is even more drastic: the non-inertial rule performs rather poorly with an R^2 of merely 0.44, while the inertial and double-inertial rules achieve R^2 values of 0.98 and 0.99. Clearly, the inertial and double-inertial rules are able to substantially improve explanatory power relative to the non-inertial specification.

The performance of each rule in levels across the extended sample is visualized in Figure 1. Rules and benchmarks incorporating inertia all track the level of the federal funds rate closely. Only upon careful inspection, as highlighted in Carlstrom and Fuerst (2014), does the reduction in the phase shift of the predictions afforded by the double-inertial rule becomes visible, particularly during the hiking cycles of the mid-1990s and mid-2000s and the cutting cycle associated with the 2001 recession. We thus turn to a different measure to more clearly highlight the descriptive benefits of double-inertia over these periods.

We construct a second R^2 variant which evaluates the rules' predictions of *changes* to the policy rate, rather than its levels. In essence, this "next-quarter changes" R^2 measure, which we denote R^2_Δ , compares realized changes in the policy rate to the changes predicted by each rule. The predicted adjustment is taken to be the difference between

Table 2: Model Fit Statistics by Sample

Policy Rule	Pre-2007					Post-1987				
	R^2	R^2_{Δ}	R^2_4	AIC	BIC	R^2	R^2_{Δ}	R^2_4	AIC	BIC
Non-inertial	0.78	-3.62	0.78	230.23	239.61	0.44	-17.44	0.44	559.06	571.15
Inertial	0.97	0.30	0.80	86.55	98.27	0.98	0.27	0.82	160.44	175.56
Double-inertial	0.98	0.59	0.81	47.93	61.99	0.99	0.57	0.83	100.46	118.60
AR(1)	0.95	0.02	0.54	109.09	116.12	0.97	-0.07	0.65	215.80	224.87
AR(2)	0.98	0.53	0.70	53.70	63.07	0.98	0.50	0.76	126.51	138.61

Note: Predictions below 0 are truncated for the calculation of the goodness of fit measures. R^2 measures fit for 'next-quarter levels' of the FFR, R^2_{Δ} measures fit for 'next-quarter changes', and R^2_4 measures fit for year-ahead levels. AIC and BIC are computed from the model log-likelihoods. The 'Pre-2007' sample runs from 1987:Q4 to 2006:Q4, and is estimated with OLS. The full 'Post-1987' sample runs from 1987:Q4 to 2025:Q3, and is estimated with Tobit regression.

the rule-implied rate and the lagged realized policy rate. Formally,

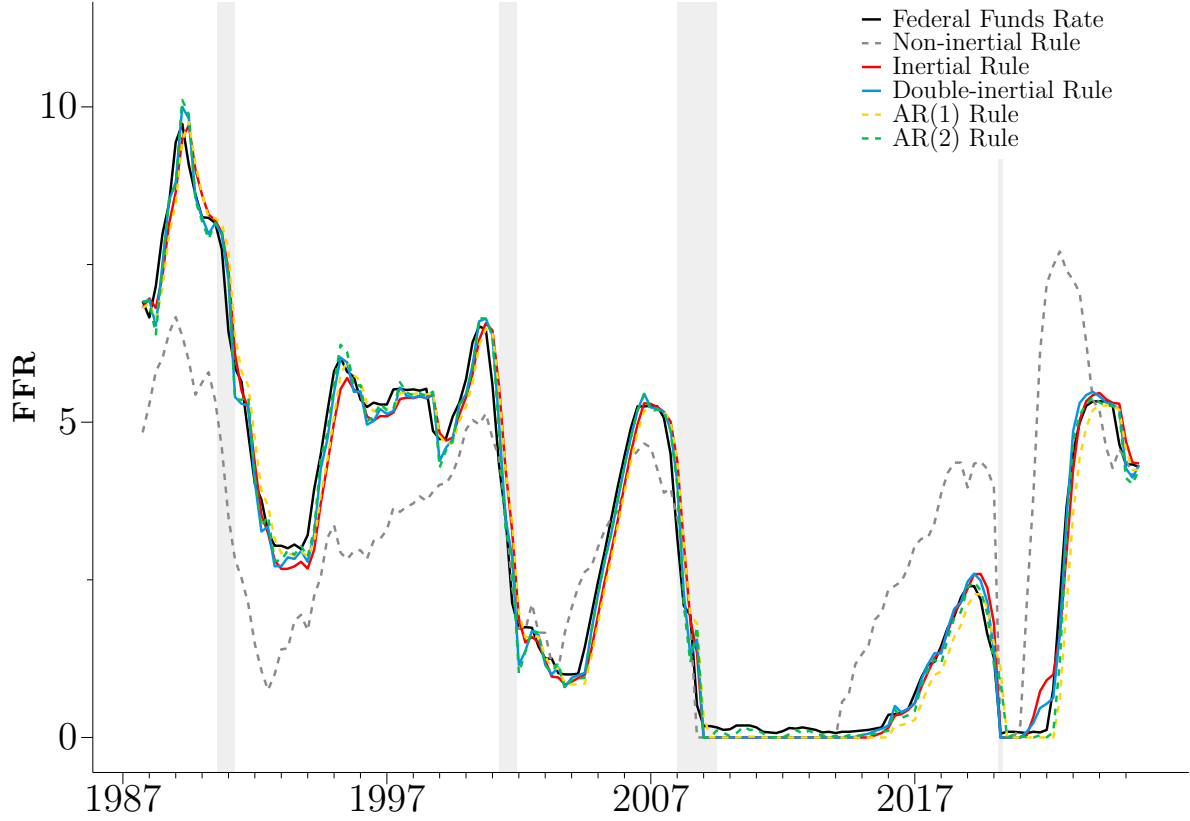
$$R^2_{\Delta r} = 1 - \frac{\sum_t \left[(i_t - i_{t-1}) - (\hat{i}_{t,r} - i_{t-1}) \right]^2}{\sum_t \left[(i_t - i_{t-1}) - \bar{\Delta i} \right]^2}, \quad (10)$$

where $\bar{\Delta i}$ is the average change in the federal funds rate over the sample. Because this measure evaluates predicted changes relative to the lagged realized policy rate, and the non-inertial rule is not anchored to that value, its performance is far worse than a rule which simply takes the average change as its prediction in each period.

The resulting values of the next-quarter changes R^2 measure are reported in the second and seventh columns of Table 2. In the pre-2007 sample, the explanatory power of the estimated inertial and double-inertial rules differs markedly. The inertial rule explains only 30 percent of the variation in changes to the federal funds rate, while the double-inertial rule explains 59 percent, corresponding to an increase of 29 percentage points in explanatory power. The same pattern appears in the full sample, where the inertial rule explains 27 percent of the variation in policy rate adjustments and the double-inertial rule explains 57 percent. That the double-inertial rule is able to explain more than twice the variation in changes to the policy rate than the inertial rule can strongly supports our contention that that gradualism in monetary policy encompasses not only persistence in the level of the policy rate, but a meaningful level of persistence in the *pace* of policy adjustments.

The implications of this finding are substantial. Imagine a policymaker who, given the current stance of policy, is confined to following a rule to decide how to adjust the federal funds rate in the next period (as policymakers so often are in our structural models). Using a conventional inertial Taylor rule, the implied adjustments would bear only a limited resemblance to actual policy. By contrast, augmenting the rule with a parameter

Figure 1: Comparison of Next-quarter Levels Predictions for Estimated Policy Rules and Benchmarks



Note: This figure displays the truncated, next-quarter level predictions of the relevant estimated policy rules over their training window. These predictions are generated using realized values of the inflation and unemployment gaps, as well as realized lags of the policy rate.

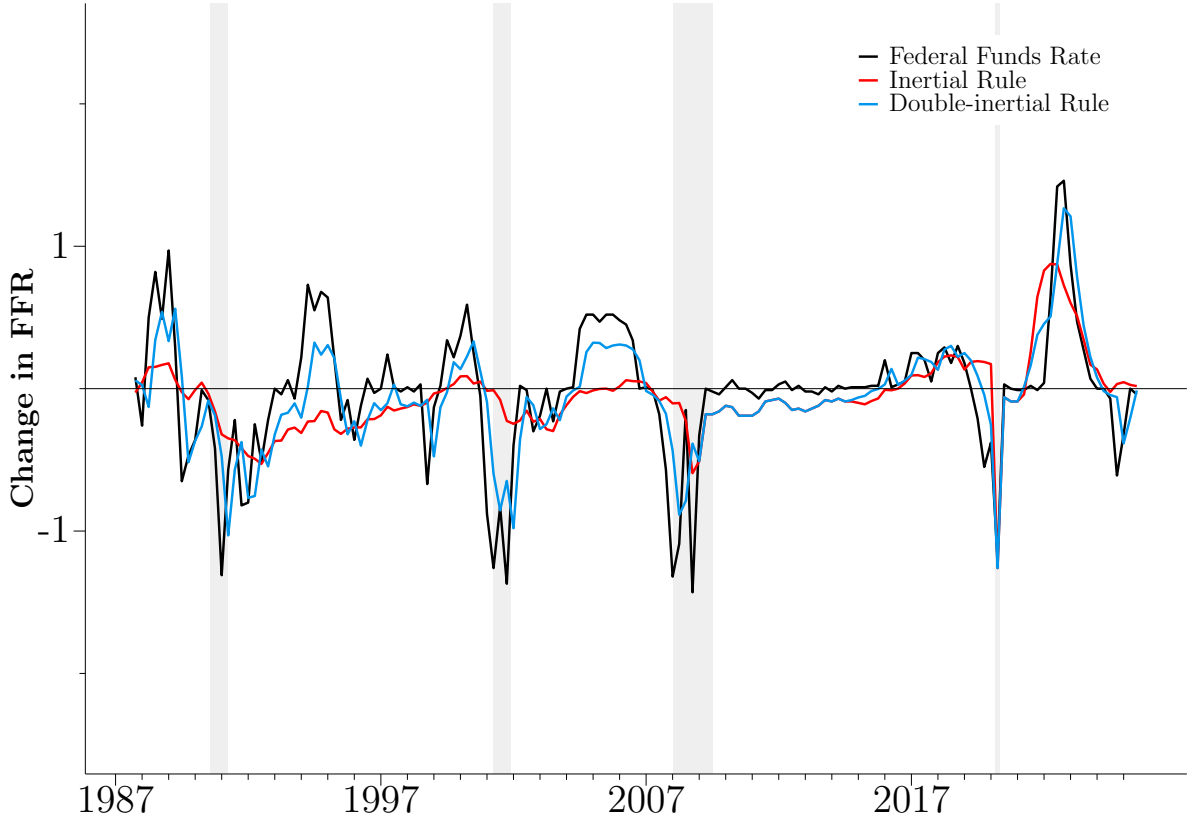
Sources: FRED.

capturing persistence in policy changes dramatically improves its ability to reproduce the actual pattern of adjustments. In effect, the double-inertial rule captures the important tendency of policymakers to adjust the policy rate in the same direction and with a similar magnitude across successive meetings.

A visualization of this hypothetical over the extended post-1987 period is provided in Figure 2. The path of adjustments prescribed by the double-inertial rule closely tracks the realized adjustments to the policy rate, albeit with some delay and often with a moderate reduction in magnitude. The inertial rule, by contrast, captures little more than broad directional shifts in policy. An intuitive explanation follows easily from these results. In first-differences, the inertial rule is no longer guided by the past: movements are based on the deviation of the Taylor rule target from the policy rate, which strictly varies with economic fundamentals. The double-inertial rule, however, uses the previous adjustment as an anchor, effectively accelerating policy adjustments when the previous adjustment indicates the need for urgency, and curtailing them otherwise.

Our final R^2 variant is designed to gauge the longer-term predictive power of the

Figure 2: Comparison of Next-Quarter Change Predictions for Estimated Policy Rules



Note: This figure displays the truncated, next-quarter change predictions of the relevant estimated policy rules over their training window. These predictions are generated- using realized values of the inflation and unemployment gaps, as well as realized lags of the policy rate.

Sources: FRED.

estimated policy rules at the one-year horizon. To calculate this “year-ahead levels” R-squared for a particular rule, we recursively generate four consecutive predictions from that rule using the data values of inflation and the unemployment rate gap as rule inputs, but where any lagged levels of the federal funds rate are taken from the previous rule prediction instead of the realized policy rate. We then evaluate the goodness of fit of the resulting four-quarter ahead prediction as:

$$R_{4,r}^2 = 1 - \frac{\sum_t (i_t - \hat{i}_{t|t-4,r})^2}{\sum_t (i_t - \bar{i})^2}, \quad (11)$$

where the prediction is defined recursively as:

$$\hat{i}_{t+s|t,r} = \begin{cases} \max\{\hat{i}, \hat{\theta}_0 + \hat{\theta}_1\pi_{t+s} + \hat{\theta}_2(u_{t+s} - \tilde{u}_{t+s}) + \hat{\theta}_3\hat{i}_{t+s-1|t,r} + \hat{\theta}_4\hat{i}_{t+s-2|t,r}\} & \text{for } s \geq 0 \\ \hat{i}_{t+s} & \text{for } s < 0 \end{cases} \quad (12)$$

The one-year-ahead R-squared metric provides a more policy-relevant assessment of rule performance by substantially reducing the mechanical dependence on interest rate predictions that arises from the inclusion of lagged interest rate terms in the estimated rules. While the recursive predictions continue to condition on observed realizations of inflation and unemployment—and thus does not constitute a true unconditional forecast of the policy rate—the resulting R-squared measure evaluates how well the rule would characterize policy decisions if a policymaker were to adhere consistently to the rule’s prescriptions over a four-quarter period, allowing the rule itself to generate the path of interest rates endogenously in response to evolving economic conditions. This approach more closely approximates the manner in which policy rules would be employed as a systematic approach over multiple decision points, rather than as single-period adjustments.

The third and eighth columns in Table 2 report the measures under each rule. Note first that, because non-inertial Taylor rules do not use lagged values of the policy rate to generate predictions, their predictions and R^2 values are unchanged from the simple next-quarter level R^2 values. They therefore serve as a useful benchmark against which to evaluate the deterioration in performance of the rules with inertial components: at 0.80 and 0.82 respectively, the explanatory power of the inertial rules falls by 16 to 17 percentage points in both subsets relative to the next-quarter levels predictions. The deterioration in explanatory power is similar for the double-inertial rules. These results highlight how imprecision surrounding the policy rate realizations compounds over time: four period lagged realizations of the policy rate are substantially less correlated with current period settings than the most recent lag.

Table 2 also reports model log-likelihood based AIC and BIC values. Both information criteria penalize the size of the model, imposing a tradeoff between goodness of fit and parsimony in selecting the best model. In both the pre-2007 and full samples, the double-inertial model is the clear winner, with substantially lower AIC and BIC values than its single-inertial counterpart. This additional evidence corroborates similar analysis by Coibion and Gorodnichenko (2012), and supports our claim that double-inertia is an important component of monetary policy and should be included in Taylor rule specifications.

4.3 Comparison to autoregressive benchmarks

The predictive performance of single- and double-inertial rules may simply come from the fact that the best prediction for tomorrow’s level of interest rates is today’s level. It is well-known that low-order autoregressive models often exhibit strikingly good forecast performance. To evaluate this view, we compare the inertial and double-inertial rules to simple AR(1) and AR(2) benchmarks.

In Table 2, the AR(1) and AR(2) benchmarks perform virtually as well as the inertial and double-inertial Taylor rules in terms of the next-quarter levels R^2 measure. The AR(2) benchmark, in particular, essentially matches the fit of the double-inertial rule and even slightly outperforms the inertial rule, achieving an R^2 of 0.98 in both the pre-2007 sample and the full sample. Remarkably, this implies that the second autoregressive term is more useful for next-quarter predictions than the inclusion of economic fundamentals, highlighting the quantitative importance of inertia in the rate of change of the policy rate.

In next-quarter changes, the relative advantage of the AR(2) benchmark over the inertial Taylor rule is further emphasized. In the full sample, the AR(2) benchmark explains 50 percent of the variation in changes to the federal funds rate, corresponding to a 23 percentage point advantage, which is almost twice the magnitude of explanatory power retained by the inertial rule in this space. This AR(2) benchmark, based solely on recent realizations of the policy rate, substantially outperforms an inertial rule grounded in economic fundamentals.

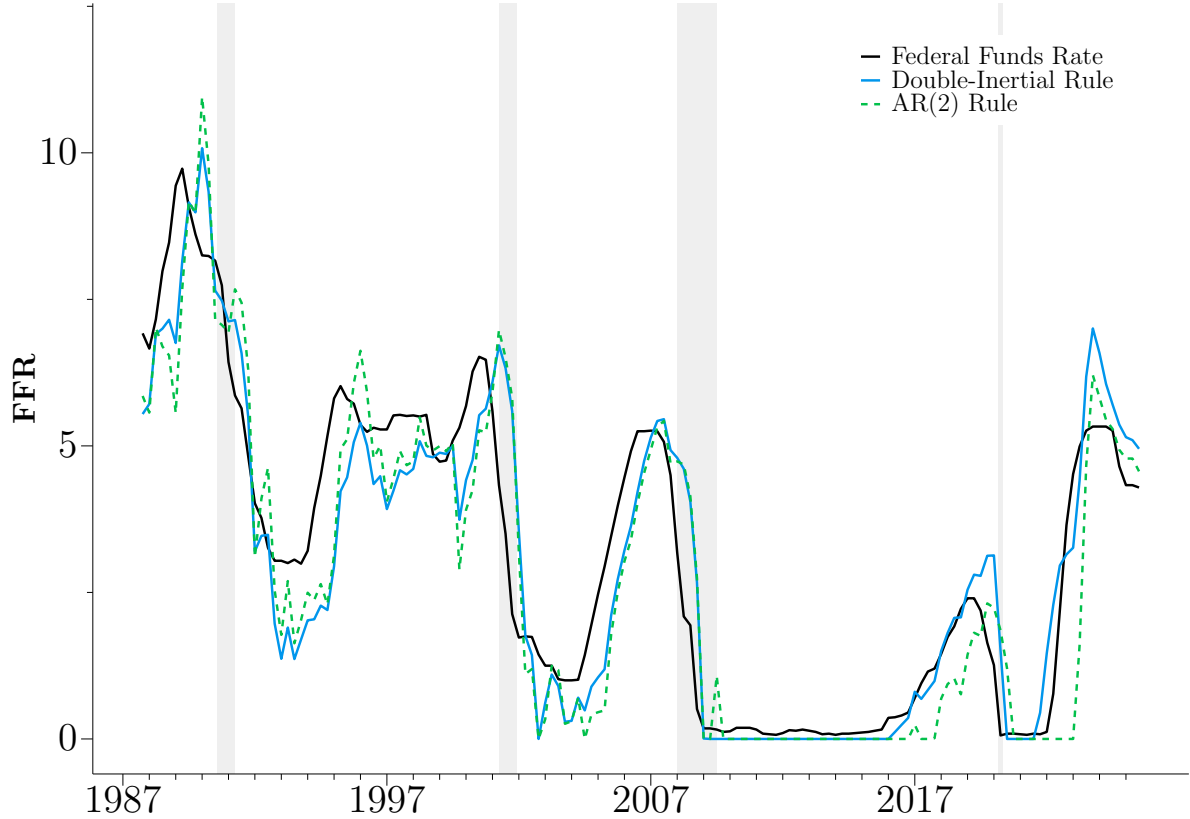
The values of the AIC and BIC criteria for the AR(2) benchmark further support this implication. In both samples, these criteria suggest that the parsimonious AR(2) specification outperforms the standard-inertial rule. However, the double-inertial rule retains a slight advantage over the AR(2) benchmark, particularly in the full post-1987 sample, confirming that it provides the best overall fit among the models tested.

While it is true that inflation and the unemployment gap do not contribute meaningful additional predictive power given the lags of the policy rate, the value of an interest rate rule is not strictly in short-term forecasts. Interest rate rules are meant to describe the actual behavior of policymakers, and to be interpreted in a context where the rule is in place over a longer period; the interest is in likely *paths* of the policy rate, and how they might change with new economic developments. Unsurprisingly, when we observe the course and direction of our Taylor rules and the settings of the policy rate over a longer horizon, the importance of economic fundamentals in guiding policymaking becomes far more evident.

From the third and eighth columns of Table 2, it can be seen that the one-year-ahead level R-squared measure for the AR(2) rule falls to 0.70 in the pre-2007 sample and 0.76 in the full sample, both of which are substantially lower than those of the inertial and double-inertial rules. Over the one-year horizon, a forecaster would be much better off assuming that policymakers follow an inertial Taylor rule than by generating forecasts based only on recent realizations of the policy rate. In other words, at the one-year horizon, the contribution of economic fundamentals to policymaking appears to outweigh that of double-inertia. This result is both intuitive and reassuring: economic conditions should be the primary guide for the path of policy, while the influence of gradualism naturally diminishes over longer horizons.

Another way to view the greater importance of economic fundamentals over longer horizons is to compare the AR(2) benchmark with the double-inertial rule. As illustrated in Figure 3, neither rule perfectly tracks the realized policy rate at the one-year horizon, and both exhibit some degree of phase shift relative to the realized path. However, their difference becomes particularly apparent following liftoff from the effective lower bound. Through the late 2010s and early 2020s, the prescriptions of the double-inertial rule move

Figure 3: Comparison of Year-ahead Predictions for Double-Inertial and AR(2) Rules



Note: This figure displays the truncated, one-year ahead predictions of the relevant estimated policy rules over their training window. These predictions are generated using realized values of the inflation and unemployment gaps and recursive predictions of the policy rate.

Source: FRED.

almost contemporaneously with the realized path of policy, while the AR(2) rule responds with over a year of delay. This observation is in line with the AIC and BIC values across the samples, which suggest that the double-inertial model's advantage over the AR(2) benchmark is primarily concentrated in the post-2007 period.

4.4 Policy inertia before gradualism

The evidence presented thus far clearly supports our contention that double-inertia is an important force in monetary policy. The similarity of our results across both the pre-2007 sample and the full post-1987 sample serves both to corroborate prior work on double-inertia and to underscore its continued relevance through recent decades. That said, the motivation for our interpretation of these results through the lens of monetary policy gradualism remains relatively opaque. It seems entirely conceivable that the statistical significance and quantitative import of γ in our estimation does not simply reflect an inherent desire of policymakers to smooth the pace of change to the policy rate. Instead, our results could just reflect the nature of persistence in economic shocks or some other

omitted expectational dynamics, as in Rudebusch (2002, 2006).

In addressing this concern, it is useful for us to briefly revisit Coibion and Gorodnichenko (2012)’s forceful defense of interest smoothing as the primary source of policy inertia. They consider several empirical strategies in evaluating the merit of Rudebusch’s critique. First, they estimate nested Taylor rule specifications that allow for both interest-smoothing and persistent shocks. Model selection criteria favored a specification with second-order interest smoothing—just like our double-inertial rule—in which the autoregressive parameters in the error term lose their persistence. Second, the authors argue that if persistent shocks were the true source of inertia, policy responses should not themselves display persistence once conditioned on non-monetary shocks that are not themselves persistent. Using an instrumental variables strategy with non-monetary shocks as instruments, the authors find that estimates of interest smoothing remain large, supporting the interest smoothing interpretation. Third, Coibion and Gorodnichenko (2012) address Rudebusch’s argument that highly persistent monetary shocks would imply high forecastability of future interest rate changes for markets. They suggest instead that the apparent lack of forecastability reflects informational constraints, not a lack of persistence, and use Greenbook forecasts to show that future rate changes are in fact highly predictable given sufficient information, at least in the earlier samples Rudebusch analyzes. Finally, they present narrative evidence from policymakers and demonstrate robustness to a wide range of omitted variables, consistently finding strong evidence of interest rate smoothing.

With Coibion and Gorodnichenko (2012)’s forceful defense of policy inertia as a product of interest smoothing as motivation, we turn to our third primary data sample, which runs from 1960:Q1 through Volcker’s resignation in 1987:Q3. We argue that the conspicuous absence of evidence for double-inertia as a component of the monetary policy reaction function over this period provides further evidence for our interpretation of change-smoothing in the context of monetary policy gradualism.

We report the estimated parameters for each of our Taylor rules in the pre-1987 sample in Table 3. The estimated parameters are generally in line with our expectations. With the exception of the unemployment gap coefficient for the non-inertial rule, the estimates imply a policy reaction function that responds to economic developments in the anticipated direction and with reasonable magnitudes. Note that the entirety of Volcker’s tenure is included in the sample, which helps to elevate the inflation coefficients above the Taylor Principal threshold of 1. Note also that, while the natural rate estimates are somewhat smaller than expected, they also tend to be relatively imprecisely estimated, which is a product of the volatility in interest rates and inflation over the period.

The estimated value of the double-inertial coefficient, γ , contrasts sharply with the estimates acquired in the more recent samples. At 0.17, the coefficient has less than a third of the magnitude of its equivalent in both post-1987 samples, suggesting a vastly reduced persistence in policy rate changes over the period. Further, given the associated uncertainty estimate, we fail (at the 5 percent level) to reject the null hypothesis that the double-inertia parameter is equal to zero. We note, however, the maintained evidence for strong persistence in the level of the policy rate, as indicated by the estimates of ρ . This result is at odds with the standard intuition that assigns monetary policy gradualism

Table 3: Recovered Rule Parameters, Pre-1987 Sample

Parameter	Non-inertial	Inertial	Double-inertial
$\hat{r}^*$	2.23 (0.40)	2.12 (1.34)	2.26 (1.05)
$\hat{\beta}_\pi$	1.15 (0.21)	1.48 (0.63)	1.31 (0.57)
$\hat{\beta}_u$	0.03 (0.25)	-2.17 (1.38)	-1.25 (0.73)
$\hat{\rho}$		0.90 (0.05)	0.88 (0.04)
$\hat{\gamma}$			0.17 (0.11)
N	111	111	111

Note: Estimated using quarterly data from FRED. Parameters $\hat{r}^*$, $\hat{\beta}_\pi$, $\hat{\beta}_u$, $\hat{\rho}$, and $\hat{\gamma}$ are recovered from regression coefficients, and Newey-West standard errors with a lag length of 4 are computed via the delta method. The 'Pre-1987' sample runs from 1960:Q1 to 1987:Q3, and is estimated with OLS.

only to the recent period, and may provide support for the argument that single-inertia partially reflects persistence in economic shocks, rather than just the desire to smooth the level of the policy rate.

We next evaluate the goodness of fit for our selection of rules and benchmarks trained over the pre-1987 sample. The results are reported in 4.

The next-quarter levels R^2 estimates are in line with our expectations. Over the pre-1987 period, the performances of the inertial and double-inertial Taylor rules in levels are indistinguishable at 0.90. As in the post-1987 samples, both autoregressive benchmarks achieve impressive performance. However, across all four models with inertial components, the levels of the R^2 measure are noticeably lower relative to the previously analyzed samples, owing to the less disciplined movements of the federal funds rate during earlier policy regimes.

In contrast to the estimates for the post-1987 samples, the story does not change with the next-quarter changes R^2 measure. The double-inertial rule held a 30 percentage point advantage in explanatory power over the inertial rule in the full post-1987 sample, more than doubling the proportion of variation in changes to the federal funds rate that could be explained. In the pre-1987 subset, however, the advantage drops to merely 3 percent. Further, the levels of the measure for both the inertial and double-inertial rules are substantially reduced relative to the corresponding measures in the later sample, falling

Table 4: Model Fit Statistics, Pre-1987 Sample

Policy Rule	R^2	R^2_{Δ}	R^2_4	AIC	BIC
Non-inertial	0.56	-2.89	0.56	516.36	527.20
Inertial	0.90	0.11	0.67	354.17	367.71
Double-inertial	0.90	0.14	0.66	353.33	369.59
AR(1)	0.89	0.03	0.57	360.15	368.28
AR(2)	0.90	0.07	0.58	357.15	367.99

Note: Predictions below 0 are truncated for the calculation of the goodness of fit measures. R^2 measures fit for 'next-quarter levels' of the FFR, R^2_{Δ} measures fit for 'next-quarter changes', and R^2_4 measures fit for year-ahead levels. AIC and BIC are computed from the model log-likelihoods. The 'Pre-1987' sample runs from 1960:Q1 to 1987:Q3, and is estimated with OLS.

from 0.27 and 0.57 to 0.11 and 0.14 respectively. These deficiencies come despite the fact that the estimated non-inertial rule actually held slightly more explanatory power over the earlier sample, suggesting that there is virtually no role for double-inertia in explaining policy rate adjustments.

This conclusion is reinforced by the values of the year-ahead levels R^2 statistic. At this horizon, the double-inertial rule is actually *outperformed* by the standard inertial rule. In other words, when evaluated over longer horizons, incorporating double-inertia into the policy reaction function does not improve the model's ability to track the realized path of the federal funds rate in the pre-1987 sample. If anything, the additional persistence implied by the double-inertial specification appears to fit more to noise than to signal, slightly worsening the performance relative to the simpler inertial rule.

Unsurprisingly, these findings materialize in the AIC and BIC values as well. For the BIC criterion in particular, which tends to favor more parsimonious models than the AIC criterion, the double-inertial rule is actually outperformed by the inertial rule and *both* autoregressive benchmarks, though, as with the AIC, all of the numerical values cluster in a very tight range.

Taken together, these results suggest that double-inertia plays little role in describing the dynamics of policy rate adjustments prior to the late-1980s shift toward gradualism. It is only in the more recent period that the evidence for double-inertia emerges. While this contrast is consistent with a gradualist interpretation, it does not by itself rule out alternative explanations. In particular, one could argue changes in the persistence of macroeconomic shocks or in the structure of the macroeconomy could give rise to similar differences across samples. It is also possible that our results partially reflect the movement of the Federal Reserve away from monetary aggregates, or toward more forward-looking policy. Thus, our results by no means offer an irrefutable defense of the gradualist interpretation of double-inertia; however, they serve to lend credence to such

a description, especially in light of [Coibion and Gorodnichenko \(2012\)](#)’s work.

5 Robustness

In this section, we carry out an assortment of robustness exercises to ensure that our primary empirical result, namely the predictive importance of double-inertia in Taylor rules over the last few decades, is not specific to our particular data and procedures. We conduct two types of robustness checks. First, we vary the sample period across five alternative windows, with the resulting performance for each Taylor rule variant reported in Table 5. Second, we modify the data construction in three ways: using real-time data, shadow rates during ELB periods, and the output gap instead of unemployment gap, with results reported in Table 6.

The first two additional subsets that we consider simply adjust the starting date for our main samples from 1987:Q4 to 1983:Q1. This extended window captures the second half of Paul Volcker’s term as Chair of the Federal Reserve, a period associated with enhanced central bank credibility, reduced inflation expectations, and macroeconomic stability. Gradualism in the United States is sometimes interpreted as beginning during the Great Moderation, and as such we test to see if double-inertia still plays a significant role over the period. Our results strongly suggest that it retains its importance, with the double-inertial rule in both windows still able to explain more than twice the variation in changes to the policy rate than can the inertial rule.

The next subset covers exactly the extended window from the main specification, running from 1987:Q4 through 2025:Q3. However, rather than run a Tobit regression to address the censorship of the ELB, we run OLS on the subset *excluding* the ELB windows. This exercise serves to ensure that our evidence for double-inertia is not merely a product of censorship. Again, our findings prove robust, with the double-inertial rule explaining an additional 28 percentage points of variation in changes to the federal funds rate than can the single inertial rule.

The penultimate subset reports results estimated from 2007:Q1 through 2025:Q3. As previous work on double-inertia has reported primarily on the period prior to 2007, this window corresponds exclusively to the window of new data that our extended specification offers. Here, the relevance of change-smoothing in policy rate settings remains evident, as the double-inertial rule is able to explain substantially more variation in changes to the policy rate than can the single-inertial rule. However, relative to other specifications, the advantage is somewhat reduced. This is not exceptionally surprising, however, as much of the signal throughout the period is obfuscated by the ELB.

Finally, we report results for a specification including the entire sample, running from 1960:Q1 through 2025:Q3. Here, the benefits from including double-inertia in the specification are quite small. This comes as no surprise, however, as the results simply reflect how the inclusion of the pre-gradualism window in the sample dominates the positive results in recent decades. This finding once again highlights the connection between double-inertia and the timing of the gradualist regime.

For our second set of robustness exercises, we return our focus to the original sample

Table 5: Model Fit Statistics, Additional Samples

Subset	R^2 <i>Statistic</i>	Non-inertial	Inertial	Double-inertial	AR(1)	AR(2)
Post-1983, Pre-2007	R^2	0.68	0.96	0.97	0.96	0.97
	R^2_{Δ}	-5.97	0.14	0.38	0.03	0.32
	BIC	354.86	158.53	131.43	161.30	131.01
Post-1983 Tobit	R^2	0.41	0.98	0.98	0.97	0.98
	R^2_{Δ}	-20.92	0.19	0.41	-0.05	0.34
	BIC	705.41	230.39	192.83	269.88	210.03
Post-1987, No ELB	R^2	0.33	0.97	0.98	0.95	0.98
	R^2_{Δ}	-12.30	0.31	0.59	0.03	0.55
	BIC	491.50	152.39	95.99	182.89	98.28
Post-2007 Tobit	R^2	0.50	0.98	0.98	0.94	0.97
	R^2_{Δ}	-8.37	0.56	0.71	-0.22	0.48
	BIC	193.67	61.81	48.14	110.53	75.43
Full Sample Tobit	R^2	0.58	0.95	0.95	0.94	0.95
	R^2_{Δ}	-6.31	0.12	0.16	-0.02	0.06
	BIC	1140.92	624.20	623.28	667.35	655.38

Note: Predictions below 0 are truncated for the calculation of R^2 . BIC is computed from model log-likelihoods. For detailed descriptions of the relevant sample windows, please see the main text.

Table 6: Model Fit Statistics, Robustness Exercises

Exercise	Statistic	Non-inertial	Inertial	Double-inertial	AR(1)	AR(2)
Real-time Data	R^2	0.78	0.98	0.99	0.97	0.98
	R^2_{Δ}	-6.73	0.24	0.53	-0.05	0.46
	BIC	374.19	140.99	99.44	175.09	113.95
Wu-Xia Shadow Rate	R^2	0.39	0.98	0.99	0.97	0.99
	R^2_{Δ}	-20.72	0.22	0.54	0.01	0.47
	BIC	597.75	166.52	118.20	199.99	124.95
Output-gap Specification	R^2	0.47	0.98	0.99	0.97	0.98
	R^2_{Δ}	-16.52	0.32	0.59	-0.07	0.50
	BIC	575.70	170.40	115.58	224.87	138.61

Note: The samples used in the reported statistics for the Wu-Xia shadow rate and output-gap based specifications run from 1987:Q4 to 2025:Q3, in line with our full post-1987 sample reported in the main results. The real-time sample runs only from 1989:Q1 to 2020:Q4 due to data availability constraints. Note also that reported statistics for the shadow rate exercise are calculated without truncating predictions below 0. BIC is computed from model log-likelihoods.

of interest and consider several new datasets that we use to gauge the broader relevance of our results.

First, in the spirit of [Orphanides \(2001\)](#), we replace our “realized” measures of the inflation and unemployment gaps with real-time data constructed using the Philadelphia Fed’s Tealbook (formerly Greenbook) Data Set. Specifically, we use the NAIRU and UNEMP variables to construct a real-time measure of the unemployment gap, and we combine the PCPIX and PPCEX variables, stitching them together in 2000, to construct a real-time measure of inflation. These alterations eliminate potential endogeneity concerns stemming from ex-post revisions, and allow us to model policy responses as a function of the data that policymakers observed in real-time. However, because of limitations on available information in this dataset, our estimation window is restricted from 1989:Q1 to 2020:Q4.

Next, we use data on the Wu-Xia Shadow Fed Funds rate to replace our observations for the effective federal funds rate during the ELB periods in the 1987 through 2025 sample. Because the shadow rate data provides us with a plausible, uncensored proxy for the latent policy rate over the ELB episodes, we use OLS rather than Tobit regression to estimate our models. This exercise allows us to incorporate information about the latent stance of policy derived from a term-structure model and Treasury yields into our analysis.

Finally, we test the robustness of our results to an output-gap based Taylor rule specification as opposed to the unemployment-gap based specification that was at the center of our earlier results. To construct our output-gap measure, we simply calculate percent deviations between the GDPC1 and GDPOT measures from the St.Louis FRED database. This exercise serves to align our analysis more closely with the literature that focuses on output-gap based Taylor rule specifications. The results of these robustness

exercises are compiled in Table 6. Each exercise produces strong evidence in line with our earlier findings. Using real-time economic data, the inertial rule is able to explain only 24 percent of variation in changes to the federal funds rate over the period while the double-inertial rule is able to explain 53 percent. Replacing the ELB data with the Wu-Xia Shadow Rate leads to similar explanatory results, with the inertial and double-inertial rules achieving next-quarter change R^2 values of 0.22 and 0.54 respectively. The output-gap based versions of the rules both performed even better, and the double-inertia rule maintained its advantage in the next-quarter changes R^2 metric: at 0.59, the double-inertia rule explains an additional 27 percentage points of variation in changes to the federal funds rate over the relevant period as compared to the standard inertial rule. Both the quantitative significance of these results and their implications for the prevalence of double-inertia in monetary policy are nearly identical to the previous results reported for our main specification.

6 Conclusion

In this paper, we studied the descriptive performance of the double-inertial Taylor rule in characterizing U.S. monetary policy. While the standard inertial Taylor rule models policy rate settings as based on partial adjustments towards an intermediate rate target prescribed by an embedded Taylor rule, the double-inertial rule also incorporates persistence in past changes to the policy rate. In this sense, the double-inertial rule captures a more nuanced picture of how gradualism in monetary policy may operate by suggesting that policymakers are interested in smoothing both the level and the pace of change of the policy rate.

Our estimates over the post-Volcker period are consistent with a monetary policy gradualism that includes change smoothing. Double-inertia significantly improved model fit over the relevant period as compared to the single-inertial Taylor model. Quantitatively, the double-inertial model more than doubled the proportion of variation in changes to the policy rate that could be explained relative to the inertial rule. In addition, the exceptional performance of our $AR(2)$ benchmark underscores the high import of this change-smoothing behavior relative to economic fundamentals in the short term, though our findings also confirm the greater explanatory power of the latter over longer horizons. Finally, we document that evidence for double-inertia was considerably weaker in the period preceding Volcker’s resignation, a finding consistent both with structural changes in monetary policy identified in the existing literature and our interpretation of our results in the context of gradualism.

Our findings regarding the explanatory power and descriptive value of double-inertia in Taylor rules have potentially important implications for policymakers, private agents, and future research alike. By providing a more complete empirical characterization of gradualism, the double-inertial Taylor rule offers a sharper benchmark than standard inertial specifications. This points to the use of standard inertial rules in macroeconomic models as a source of misspecification. Additionally, insofar as policy rules serve as useful tools for interpreting monetary policy, this refinement may help align expectations

between policymakers and the public and motivate further study of both the positive and normative dimensions of gradualism.

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